

### C1.1 How has the Earth's atmosphere changed over time, and why?

#### Teaching and learning narrative

The Earth, its atmosphere and its oceans are made up from elements and compounds in different states. The particle model can be used to describe the states of these substances and what happens to the particles when they change state. The particle model can be represented in different ways, **but these are limited because they do not accurately represent the scale or behaviour of actual particles, they assume that particles are inelastic spheres, and they do not fully take into account the different interactions between particles.**

The formation of our early atmosphere and oceans, and the state changes involved in the water cycle, can be described using the particle model.

Explanations about how the atmosphere was formed and has changed over time are based on evidence, including the types and chemical composition of ancient rocks, and fossil evidence of early life (1aS3).

Explanations include ideas about early volcanic activity followed by cooling of the Earth resulting in formation of the oceans. The evolution of photosynthesising organisms, formation of sedimentary rocks, oil and gas, and the evolution of animals led to changes in the amounts of carbon dioxide and oxygen in the atmosphere.

#### Assessable learning outcomes

*Learners will be required to:*

1. recall and explain the main features of the particle model in terms of the states of matter and change of state, distinguishing between physical and chemical changes and recognise that the particles themselves do not have the same properties as the bulk substances
2. **explain the limitations of the particle model in relation to changes of state when particles are represented by inelastic spheres**
3. use ideas about energy transfers and the relative strength of forces between particles to explain the different temperatures at which changes of state occur
4. use data to predict states of substances under given conditions
5. interpret evidence for how it is thought the atmosphere was originally formed
6. describe how it is thought an oxygen-rich atmosphere developed over time

#### Linked learning opportunities

##### Practical work:

- measure temperature against time and plot a cooling curve for stearic acid or heating curve for ice

##### Ideas about Science:

- use the particle model to explain state changes (1aS3)
- distinguish data from explanatory ideas in accounts of how the atmosphere was formed (1aS3)

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| Teaching and learning narrative                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Assessable learning outcomes<br><i>Learners will be required to:</i>                                                                                                                                                                                                     |
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| <p>Our modern lifestyle has created a high demand for energy. Combustion of fossil fuels for transport and energy generation leads to emissions of pollutants.</p> <p>Carbon monoxide, sulfur dioxide, nitrogen oxides and particulates directly harm human health. Some pollutants cause indirect problems to humans and the environment by the formation of acid rain and smog. Scientists monitor the concentration of these pollutants in the atmosphere and strive to develop approaches to maintaining air quality (IaS4).</p> <p>The combustion reactions of fuels and the formation of pollutants can be represented using word and symbol equations. The formulae involved in these reactions can be represented by models, diagrams or written formulae. The equations should be balanced.</p> <p>When a substance chemically combines with oxygen it is an example of oxidation. Combustion reactions are therefore oxidation.</p> <p>Some gases involved in combustion reactions can be identified by their chemical reactions.</p> | 7. describe the major sources of carbon monoxide and particulates (incomplete combustion), sulfur dioxide (combustion of sulfur impurities in fuels), oxides of nitrogen (oxidation of nitrogen at high temperatures and further oxidation in the air)                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 8. explain the problems caused by increased amounts of these substances and describe approaches to decreasing the emissions of these substances into the atmosphere including the use of catalytic converters, low sulfur petrol and gas scrubbers to decrease emissions |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 9. use chemical symbols to write the formulae of elements and simple covalent compounds                                                                                                                                                                                  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 10. use the names and symbols of common elements and compounds and the principle of conservation of mass to write formulae and balanced chemical equations                                                                                                               |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 11. use arithmetic computations and ratios when balancing equations<br>M1a, M1c                                                                                                                                                                                          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 12. describe tests to identify oxygen, hydrogen and carbon dioxide<br>PAG2                                                                                                                                                                                               |
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#### **Linked learning opportunities**

##### **Ideas about Science:**

- unintended impacts of burning fossil fuels on air quality (IaS4)
- catalytic converters, low sulfur petrol and gas scrubbers as positive applications of science (IaS4)

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